Name _____

Company Background

Founded in 1989, US Foods is a leading American foodservice distributor. US Foods offers more than 350,000 national brand products and its own "exclusive brand" items, ranging from fresh meats and produce to prepackaged and frozen foods. The company employs approximately 25,200 people in more than 60 locations nationwide and provides food and related products to more than 250,000 customers, including independent and multi-unit restaurants, healthcare and hospitality entities, government and educational institutions. The company is based in Rosemont, Illinois.

Note that US Foods' fiscal year ends for 2023 and 2022 occur on December 30, 2023 and December 31, 2022, respectively.

Part I: US Foods– General (30 points)

Ignore any tax effects when answering questions below.

1. (2 points) What is US Foods' total current assets as of December 30, 2023?

   b/s: "Total current Assets" = 4,031.

   * not 13,187 b/c that's "Total Assets"

2. (2 points) What is US Foods' total shareholders' equity as of December 30, 2023?

   b/s: "Total shareholder's Equity" = 4,749.

3. (2 points) What is US Foods' net income ("Net income") for the year ended December 30, 2023?

   IS: "Net income": 506.

4. (3 points) US Foods' market value on December 30, 2023 was $11,488 million. What is one reason why market equity might differ from shareholders' equity?

   growth prospects. human capital. (new CEO; ex: starbucks).

   → remember market equity (# shares × $/share) is relatively subjective. Stocks move up and down based on prospects and potential future cash flows.

Name _____

5. Investors use return-on-assets (ROA = Net Income / Total Assets) to evaluate a company's profitability and understand the return earned on each dollar that firm invests in its assets. Ignore any tax effects.

(a) (3 points) What is US Foods' ROA for fiscal year ended December 30, 2023?

ROA = 506 ("Net Income" on IS) / 13,187 ("Total Assets" on B/S). ≈ 3.84% or 0.0384.

(b) (9 points) For each of the transactions listed below, indicate its effect on US Foods' ROA. Would ROA have been higher than, lower than or the same ROA as what you computed in part (a) if US Foods had engaged in the transaction?

i. A customer paid its outstanding balance with US Foods that was previously recorded as accounts receivable.

Higher Lower The same

ROA = NI / Total Assets

DR Cash (+A). CR A/R (−A).

No effect on NI b/c A/R was already accounted for initially.
No effect on Assets b/c adding cash but subtracting A/R (both assets)

ii. US Foods repaid a loan of $100 million to the bank on January 1, 2023.

Higher Lower The same

ROA = NI / TA↓

DR loan (−L) 100m.
CR cash (−A) 100m.

No effect on NI b/c no sales have been performed.
Total Assets decreases b/c we pay out cash (asset).

Dividing by a smaller number results in higher output.

iii. US Foods sold additional goods at a profit.

Higher Lower The same

CR Revenue (+R)
DR Cash (+A).
CR Inventory (−A).
DR COGS (+E).

since it's sold "at a profit", the magnitude of our increase in cash is more than our decrease in cash.

ROA = NI↑ / TA↓

Dividing by a smaller number AND numbers being divided is larger.

Name _____

means creating an asset.

(c) (9 points) In the fiscal year ended December 30, 2023, US Food's internal auditing team discovered that one of US Food's segments had improperly capitalized $5 million in R&D payments (i.e., they created an R&D asset instead of directly expensing the R&D payments). The auditors corrected the error. Had the auditors not corrected the error, would the following financial statement items be understated, overstated, or accurate? Please circle one.

what if (capitalize) - used cash (CR cash $5m) for an asset (DR Asset $5m)

Real world (expense): used cash (CR cash $5m) for an expense (DR R&D Expense $5m).

i. Total assets as of December 30, 2023?

Understated Overstated Accurate

overstated b/c the R&D was improperly capitalized. An asset was created on accident.

ii. Net income in the fiscal year ended December 30, 2023?

Understated Overstated Accurate

NI = Revenue − Expense

→ if there is no R&D, we are not subtracting something that needs to be subtracted. NI is overstated.

iii. Retained earnings in the fiscal year ended in December 30, 2023?

Understated Overstated Accurate

$RE = RE_{t-1} + NI - \text{Dividends}$

this Net Income piece is also included in the calculation of Retained Earnings.

b/c NI is overstated, RE is also overstated (since we're not subtracting expense).

Name _____

Part II: US Foods – Cash Flows (20 points)

The following questions pertain to US Foods' Statement of Cash Flows for the year ended December 30, 2023.

going off based on cash receipts. reconciling from net income.

1. (6 points) If US Foods had used the direct rather than indirect method on its Statements of Cash Flows, would the following items be higher, lower or the same?

   (i) Total Cash Flows

   Higher Lower The same

   (ii) Cash flows from operations

   Higher Lower The same

   (iii) Cash flows from investing activities

   Higher Lower The same

* Lecture 2, page 87.
→ "Both methods result in the same of "Cash from Operations".
→ "CFI and CFF are computed identically under the two methods".

If you think about it, why would those be different? Obviously, more cash is good. If one method resulted in more cash in the books, every company would use that method.

2. (3 points) US Foods paid interest in 2023. How are these interest payments classified? Circle one.

   Cash Flows from Operations Cash Flows from Investing Cash Flows from Financing

this one is just straight up knowing this knowledge. I thought about it like this: every company has a bunch of loans. those loans have consistent interest payments. B/c it's so commonly occurring (monthly), it's classified under Operations.

Name _____

3. (3 points) The cash flow statement shows that US Foods recorded a gain on disposal of property and equipment of $6 million. How are the cash proceeds from disposal of this property and equipment classified? Circle one.

Cash Flows from Operations | Cash Flows from Investing | Cash Flows from Financing

PP&E is always an investment. PP&E includes things like buildings, machines, factories, etc.

4. (2 points) The cash flow statement shows that US Foods recorded depreciation and amortization of $395 million. Did this amount have a positive, negative, or no impact on cash flows from operating activities? Circle one.

Positive | Negative | No impact

D&A is a non-cash expense. Just like a factory is decreasing in value based on how old it is, doesn't mean the company has to pay cash. D&A is just a way for companies to keep track of the long term assets' value.

5. (6 points) Circle all items that represent a source of cash flow for US Foods:

Cash Flows from Operations | Cash Flows from Investing | Cash Flows from Financing

this is positive.

these are negative.

Name _____

Part III: US Foods – Accounts Receivable (30 points)

The following questions relate to US Foods' accounts receivable (Please only consider "Accounts receivable," not "Vendor receivables."). Ignore any tax effects when answering questions below.

1. (3 points) How much do customers owe US Foods as of December 30, 2023?

→ Just b/c US foods has an allowance doesn't mean customers don't owe them for that much.

1854 + 18 = 1872. (from B/S).

2. (3 points) What percentage of receivables does US Foods NOT expect to collect as of December 30, 2023?

allow / Gross A/R = 18/1872 ≈ 0.96%.

3. (6 points) What is your best estimate of the amount of cash that US Foods collected from customers during fiscal year 2023? Assume that all sales are made on credit. → credit sales.

Gross A/R	
BB	
Credit sales	cash collected write-offs
EB	

Lecture 4, page 51.

* You absolutely need to know this T-table.

Gross A/R.	
BB ① 1735	
Sales ② 35,597	X cash collections ③ 36 write-offs.
EB @ 1872.	

① B/S, net A/R + allowance (2022)

② IS, net sales

③ Note 5, "customer accounts written off"

④ B/S, net A/R + allowance. (2023).

1735 + 35,597 − X − 36 = 1872.

∴ Algebra...

X = 35,424.

Name _____

4. (9 points) Suppose you are an analyst following US Foods. Based on your analysis of US Foods' customers and industry competitors, you think that US Foods was too optimistic about its repayment rate for credit sales made in 2023. You believe the bad debt expense in 2023 should be $5 million higher. What adjustments would you make to the following items in US Foods' financial statements? Please indicate the direction and magnitude of the adjustment, if any. Ignore any tax effects.

(i) Net income for the fiscal year ending December 30, 2023?

Higher Lower The same

If higher/lower, by how much? $5m.

→ Remember, NI = Revenue − Expenses. If Bad Debt Expense is $5m higher, NI should be lower by $5m.

(ii) Gross accounts receivable for the fiscal year ending on December 30, 2023?

Higher Lower The same

If higher/lower, by how much? _____

Gross A/R = Net A/R + Allowance.

→ whether you're accounting for uncollectible accounts in that period or ahead of time, what you are owed from customers is the exact same.

(iii) Total assets on December 30, 2023?

Higher Lower The same

If higher/lower, by how much? $5m.

→ On the b/s, you can see that allowance is not included in the calculation of Total Assets. If you increase allowance by $5m, that is taken out of the calculation of net A/R and therefore Total Assets.

Name _____

5. (9 points) Suppose that one of US Foods' international competitors uses the specific write-off method (also known as the direct write-off method) of accounting for uncollectible accounts. As an analyst, you want to compare US Foods' financials to those of its competitors as if US Foods also used the direct write-off method to account for uncollectible accounts. You should disregard the previous question's proposed change.

Please indicate the magnitude and direction of the adjustments needed for each of the following of US Foods' financial statement items. Ignore any tax effects.

(i) Net income for fiscal year ending on December 30, 2023?

Higher Lower The same

If higher/lower, by how much? 36 − 24 = $12m.

Allowance balance: BB in 2023 is 30. } 24 is added (as anticipation for the year).
EB in 2023 is 18. } (36) is subtracted as an actual charge that happened.
→ means $36m was uncollectible (written off).

Under direct write-off: 36k was not collected. However, 24k was added. (b/c we don't have that allowance anymore).

(ii) Gross accounts receivable for the fiscal year ending on December 30, 2023?

Higher Lower The same

If higher/lower, by how much? _____

gross A/R = net A/R + allowance
- only exists for allowance method.
- specific write off method doesn't have allowance. that allowance switches over to just net A/R.

(iii) Total Assets for the fiscal year ending on December 30, 2023?

Higher Lower The same

If higher/lower, by how much? $18m.

Allowance is not included in Total Assets. It's a contra-asset account (XA).
when the allowance is erased, we get back $18m.

Name _____

Part IV: US Foods – Inventories (20 points)

US Foods uses the last-in, first-out (LIFO) cost flow assumption to value its inventories. If a company uses LIFO assumption to value its inventories, US GAAP requires the company to disclose its LIFO Reserve, which is defined as:

LIFO Reserve = Inventory FIFO – Inventory LIFO

Assume a tax rate of 21%.

1. (2 points) What is the net book value of US Foods' total inventories on December 30, 2023?

B/S : "Inventories – net" = 1,600 m.

(this is the LIFO value).

2. (3 points) What is your best estimate of US Foods' inventories at the end of fiscal 2023 when valued at current replacement cost (i.e., FIFO)?

LIFO Reserve = FIFO inventory – LIFO inventory. (Lecture 5, page 29).

488 = FIFO – 1600.

(Note 2, LIFO Reserve in 2023).

FIFO = 488 + 1600 = 2,088 m.

3. (6 points) What is your best estimate of the amount of total inventories US Foods purchased in the fiscal year ending on December 30, 2023? (Hint: fill out the inventory t-account using BB, EB and COGS, and solve for Purchases.)

Inventory.	
BB	
Purchases	COGS
EB.	

* must know.
Lecture 5, page 29.

Inventory.	
BB @ 1616	
Purchases X	COGS ③ 29,449
EB ② 1600	

① B/S, inventory (2022).
② I.S., cost of goods sold (2023).
③ B/S, inventory (2023).

1616 + X – 29,449 = 1600.

... Algebra...

X = 29,433 m

Name _____

4. (9 points) Suppose that US Foods had used first-in, first-out (FIFO) as a cost flow assumption for all of their Inventories.

* what if FIFO?

(a) Would cost of goods sold be higher, lower, or the same in fiscal year 2023?

Higher Lower The same

If higher or lower, by how much? -$1m.

ALIFO reserve = LIFO COGS − FIFO COGS. (lecture 5, page 29).

488 in 2022, 488 in 2023: change of −1. (note 2).

−1 = LIFO COGS − FIFO COGS.

FIFO COGS − 1 = LIFO COGS.

FIFO COGS is higher than LIFO COGS by 1m.

(b) Would the taxes of US Foods be higher, lower, or the same for activities in fiscal year 2023?

Higher Lower The same

If higher or lower, by how much? $0.21m.

→ from part a), FIFO COGS is higher by $1m. Pretax income is lower by $1m (Revenue − COGS).

→ multiply by 0.21 (given in question): 1m · 0.21 = $0.21m.

(c) Would shareholders' equity on December 30, 2023 be higher than, lower than, or the same as the amount currently recorded? If different, by how much?

Higher Lower The same

In Increased/Decreased, by how much? _____

Retained earnings is a line item in SE.

RE = RE_{t-1} + NI − Dividends.

NI = Revenue − Expenses. → this is essentially a question about cumulative cb/c on B/S. COGS between LIFO and FIFO.

→ LIFO Reserve = Cumulative COGS LIFO − cumulative COGS FIFO.

LIFO Reserve in 2023: 488m. (this tells that cumulative COGS under LIFO was larger by 488 → NI is lower).

→ multiply by (1 − tax rate): 488 · (1 − 0.21) = $385.52. (since FIFO COGS is lower, NI is higher).

& this all rolls up back to the RE equation.